Department of Computer Science and Engineering
Bachelor of Computer Science and Engineering (BCSE)
Thesis along with Knowledge Profile (K),
Complex Engineering Problem (P), and Complex Engineering Activities (A)

- **Course Code and Title:** ART 203

- **Title of the Thesis:** The Fusion of Tradition and Modernity: Analyzing Western Culture Influence on Bengali Weddings Using Machine Learning

- **Semester:** Fall 2025

- **Student(s) Name and ID:** Al Mun Tasir Mamun (21303050)

- **Supervisor Name:** M.M Rakibul Hasan

- How are the Knowledge Profile K's (K1, K2, and K7) addressed (which are not part of P's)?

Knowledge Profile K's	**Addressed (**Explain how it was addressed?**)**
K1	Applies concepts from social and cultural sciences to interpret Bengali wedding practices as complex systems influenced by globalization. This helps frame the real-world problem in terms of cultural identity, societal impact, and continuity.
K2	Mathematical principles such as probability, statistics, and evaluation metrics (accuracy, precision, recall, F1-score) were applied in analyzing and validating the performance of machine learning models. These quantitative methods ensured reliability and accuracy of the research findings.
K7	The research directly engages with cultural and societal issues by investigating how Western influence shapes Bengali weddings. Ethical concerns, cultural sensitivity, and sustainability considerations were addressed by anonymizing data and highlighting responsible event management.

- Which P's are addressed (Requirements: P1 is must, and some or all P's from P2 to P7)?

Name of CEP (P's)	**Addressed (**Explain how it was addressed?**)**	
P1: Depth of Knowledge (Required one or more K from K3, K4, K5, K6,orK8)	**K3**	Addressed by integrating Engineering Fundamentals – data structures, algorithms.
	K4	Specialized knowledge – ML/AI
	K5	Engineering design – ML pipeline
	K6	Modern tools – Python, scikit-learn, XGBoost
	K8	Research methods – structured dataset, statistical testing.
P2: Conflicting Requirements	The work balanced accuracy versus interpretability in model selection and addressed the cultural tension between tradition and modernity. Such conflicting requirements demonstrate the challenge of finding technically sound yet socially meaningful solutions.	
P3: Depth of Analysis	Analysis of 150,000 wedding records involved building structured datasets, extracting features, and conducting classification, visualization, and cultural trend evaluation. This depth ensured both technical robustness and cultural interpretation.	
P4: Familiarity of Issues	The problem involved interdisciplinary issues across computer science, sociology, and anthropology. Addressing these areas requires familiarity with both computational methods and cultural theory.	
P5: Extent of Applicable Codes	The research followed academic codes of conduct, proper reference, and ethical data use. These ensured integrity, originality, and compliance with university guidelines.	
P6: Extent of Stakeholders	The results are relevant to academics, wedding planners, cultural organizations, and wider society by providing data-driven insights into evolving traditions. The work reflects awareness of multiple stakeholders in cultural change.	

Name of CEP (P's)	**Addressed (**Explain how it was addressed?**)**
P7: Interdependence	The project required integration of dataset preparation, preprocessing pipelines, model training, evaluation, and sociocultural interpretation. Each component was dependent on the others, reflecting real-world complexity.

- Which A's are addressed (Requirements: Some or allA1 to A5)? [Only Relates to PO(j)]

Name of CEA (A's)	**Addressed (**Explain how it was addressed?**)**
A1: Range of Resources	The thesis utilized diverse resources including large-scale datasets, ML libraries, visualization tools, and academic literature. This variety of inputs ensured comprehensive outcomes.
A2: Level of Interaction	Conducted collaboratively by two students under the supervision of a faculty member, the project involved teamwork, regular feedback, and coordination with academic mentors.
A3: Innovation	The work is innovative in combining machine learning with cultural anthropology to analyze weddings, creating a novel interdisciplinary approach. This methodology has not been widely applied before in cultural research.
A4: Consequences to Society and Environment	The findings highlight the societal consequences of cultural hybridization and the environmental implications of resource allocation in weddings. This ensures awareness of sustainability and cultural sensitivity.
A5: Familiarity	The thesis demonstrates effective communication through structured report writing, academic presentations, and clear visualization of results. It shows readiness for professional research dissemination.

Comments: This thesis successfully integrates computational techniques with cultural studies, addressing both technical and societal aspects. It demonstrates originality, ethical responsibility, and adaptability, thereby fulfilling all K, P, and A justifications required by the BCSE program.